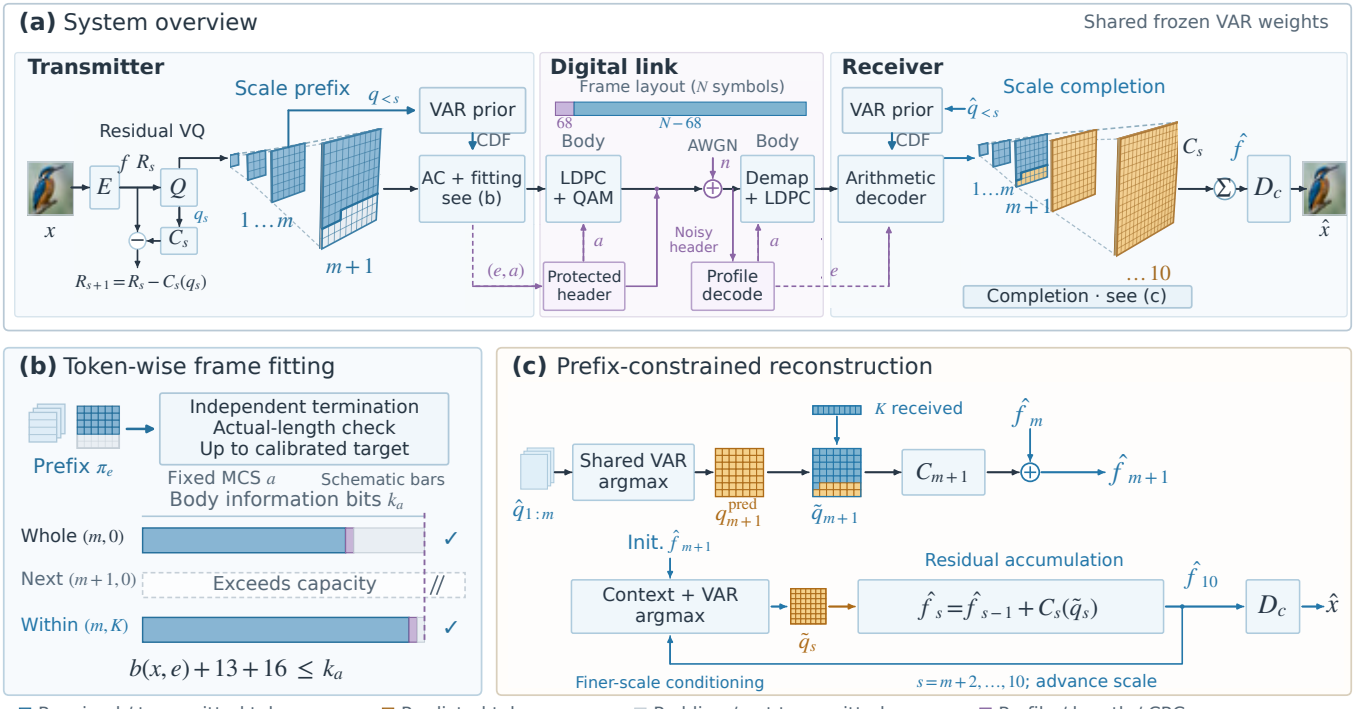

(a) System overview
Shared frozen VAR weights
Transmitter
Residual VQ
Scale prefix
VAR prior
CDF
AC + fitting see (b)
Digital link
Frame layout (N symbols)
68
N − 68
Body
AWGN
Body
LDPC + QAM
Demap + LDPC
Protected header
Noisy header
Profile decode
Receiver
VAR prior
CDF
Arithmetic decoder
Scale completion
… 10
Completion · see (c)
(b) Token-wise frame fitting
Independent termination
Actual-length check
Up to calibrated target
Prefix
Fixed MCS a
Schematic bars
Body information bits
Whole (m, 0)
Next (m+1, 0)
Exceeds capacity
Within (m, K)
b(x, e) + 13 + 16 ≤ k_a
(c) Prefix-constrained reconstruction
K received
Shared VAR argmax
Init.
Context + VAR argmax
Residual accumulation
Finer-scale conditioning
s = m+2, …, 10; advance scale
Received / transmitted tokens
Predicted tokens
Padding / not transmitted
Profile / length / CRC